MILITARY SPECIFICATION

SIMULATOR, SWITCH BOX, BOMB AND ROCKET DISPENSER SMU-64/A

This specification is mandatory for use by USAF.

1. SCOPE

1.1 This specification covers one type of simulator designated as the SMU-64/A hereafter referred to as the simulator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-I-8500	Interchangeability and Replaceability, Physical, of Component Parts for Aircraft
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery Of

STANDARDS

Military

MIL-STD-16	Electric and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation Of

DRAWINGS

66D13260	Simulator, Switch Box, Bomb and Rocket Dispenser SMU-64/A

FSC 1190

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(Copies of the above documents which are required by contractors in connection with specific procurement functions should be obtained from the Administrative Contracting Officer or as directed by the Procuring Activity.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Design and Construction. The simulator shall be designed and constructed in conformance to Drawing No. 66D13260 and subsidiary drawings.

3.2.1 Interchangeability. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable the criteria of MIL-D-1000 shall apply and new drawings shall be prepared as required. Physical interchangeability shall be in accordance with MIL-I-8500.

3.3 Performance. Satisfactory performance of the simulator shall consist of the capability to produce proper electrical simulation of PAL equipment in Nuclear Weapons.

3.3.1 Reliability. The simulator shall have a Mean Time Before Failure rating of 350 cycles of operation under service conditions.

3.4 Environmental Requirements. The simulator shall be capable of satisfactory performance during or after, as applicable, subjection to the following environmental conditions:

a. Low Temperature and Thermal Shock. The unit with test cable connected shall be capable of proper performance during and after prolonged exposure at temperature from -65°F to +165°F.

b. High Temperature and Thermal Shock. The unit with test cable connected shall be capable of proper performance during and after prolonged exposure at temperatures from +165°F to -65°F.

c. Vibration. The simulator shall be capable of satisfactory operation during and after vibration, as specified in vibration tests of this specification.

d. Humidity such as encountered in tropical areas.

e. Shock incident to transportation, handling and service conditions.

3.5 Identification of Product. The simulator, parts and assemblies shall be marked for identification in accordance with MIL-STD-130.

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3.6 Government Loaned Equipment. When so stated in the contract, the following items will be furnished the contractor, upon his request, as government loaned property for the purpose of testing the simulator:

NUMBER ITEM	PART NUMBER	DESCRIPTION OF ITEM	QUANTITY
1	5223700	Dispenser, Bomb and Rocket SUU-20/A	1
2	66D13258-7	Wiring Harness, Internal	1
3	66D13394-7	Cable Assembly, Special Purpose Electrical, Branched HRU-790/A (To be used for testing with F-111A Aircraft)	1
4	66D13392-7	Cable Assembly, Special Purpose Electrical, HRU-791/A	1
5	66D13393-7	Cable Assembly, Special Purpose, Electrical, HRU-792/A	1
6	66J13220-3	Control-Monitor DCU-116/A	1

3.7 Reference Designations. Reference designations used for marking within the unit shall be in accordance with MIL-STD-16. Markings shall not adversely affect the leakage path between conductors, or any factor of performance. Reference symbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts lists prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily viewed.

3.8 Workmanship. The simulator shall be fabricated and finished in a manner such that the criteria of appearance, fit and adherence to specified tolerances shall be observed. These criteria shall apply to all non-functioning parts as well as to functioning parts. Particular attention shall be given to marking of parts and assemblies, plating, machine screw assemblage, and freedom of parts from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests. Inspection and testing of the simulator shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 Preproduction Testing

4.2.1 Preproduction Test Sample. The preproduction test units shall be representative of the production equipment. They shall be tested by a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

4.2.2 Preproduction Test Report. In the event that preproduction testing is performed at the contractor's facility, the contractor shall, at the completion of preproduction tests, prepare a preproduction test report. The preproduction test report shall be prepared in accordance with MIL-STD-831 and three complete copies shall be furnished to the procuring activity.

4.2.3 Preproduction Tests. Preproduction tests shall consist of all tests described in "4.5 Test Methods" of the specification.

4.3 Acceptance Tests. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Tests

4.3.1 Individual Tests. Each simulator shall be subjected to the following individual tests:

a. Examination of Product

b. Conformance to Drawings

c. Functional Tests

4.3.2 Sampling Tests. Sampling tests shall consist of the following and shall be performed in the order shown:

a. Low Temperature and Thermal Shock Test

b. High Temperature and Thermal Shock Test

c. Vibration Test

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4.3.2.1 Sampling Plan. Simulators to be subjected to the sampling tests shall be selected at random from production items as follows:

a. First 100 produced, 1 sample from each lot of 20.

b. Each additional 1000 produced, 1 sample from each lot of 100.

4.3.2.2 Rejection and Retest. When a sample item fails to satisfy the specified tests, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.3.2.3 Individual Tests May Continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or produced later shall not be made until it is determined that the items meet all requirements of this specification.

4.3.3 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions. Unless otherwise specified, tests shall be performed at room atmospheric conditions, and the unit shall be at room temperature when each test is started.

4.5 Test Methods

4.5.1 Examination of Product. Each simulator shall be examined visually to determine compliance with all requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 Conformance to Drawings. Each simulator shall be inspected to determine conformance to the engineering drawings, subsidiary drawings, and all specifications and standards referenced therein. Failure to meet these requirements shall be cause for rejection.

4.5.3 Functional Tests. Functional tests shall be performed on each simulator to ensure that the proper electrical responses are indicated during the tests.

4.5.3.1 Acceptance Test Circuit. A test circuit as shown in Figure 1, shall be set up and used to perform the Functional Tests.

4.5.3.2 Test Procedures. Connect the simulator to the test set up as shown in Figure 1 and apply power to tester. This procedure will be applied to the test in the order indicated in Table 1. Failure to meet these requirements shall be cause for rejection.

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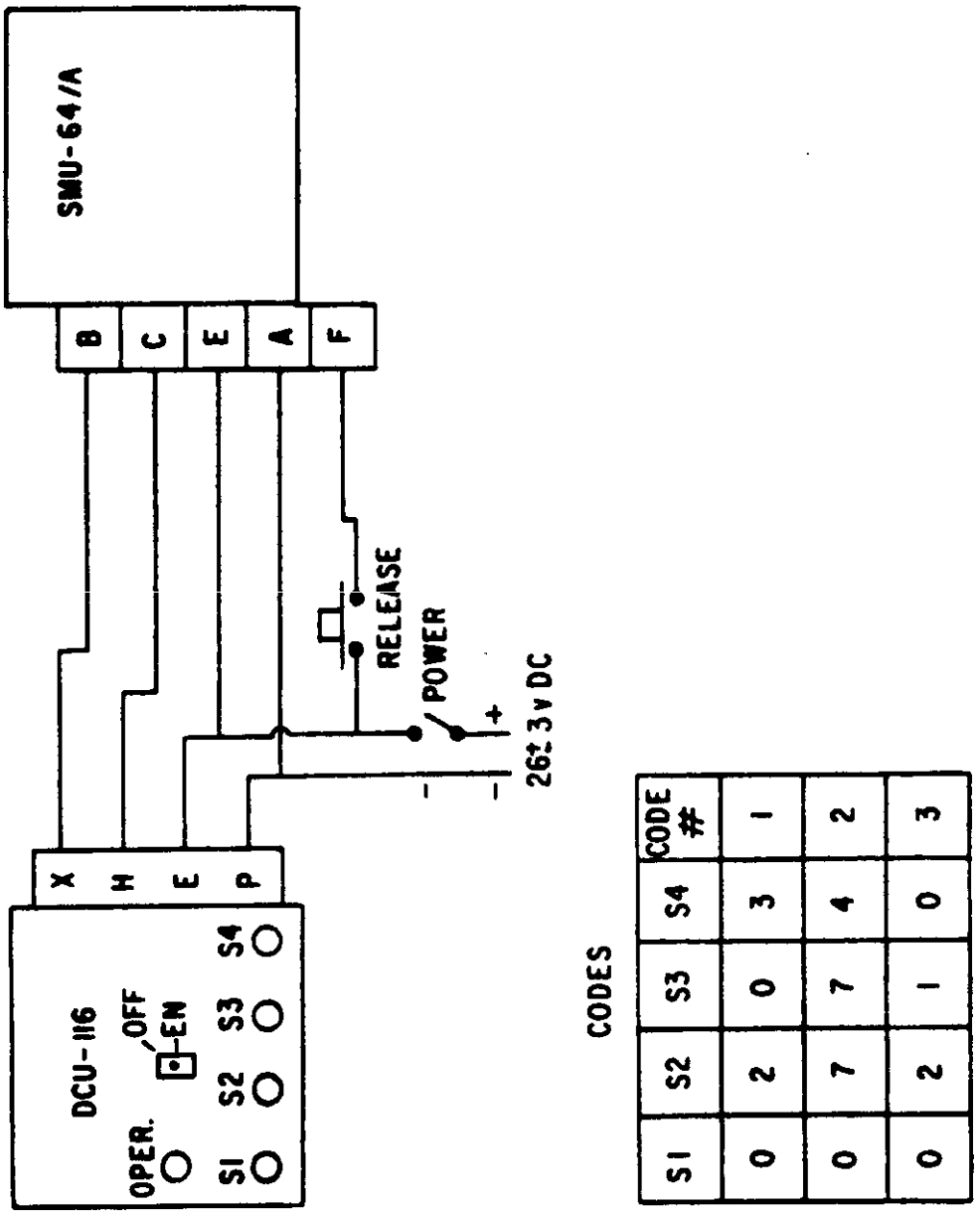


Figure 1. SMU-64/A CHECK OUT CIRCUIT AND CODES

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TABLE I

ACCEPTANCE TESTS

DCU-116 CONTROL	OPER. LT.	CENTER STATION LIGHT	COMMENTS
1. EN. Switch to "Off"		Pull hood	Power on 23 or 29 VDC
2.			Press Release Switch
3. Press Oper. Light	ON	OFF	
4. Set Code into S1, S2, S3 and S4			Codes are found on Figure 1
5. EN. Switch to "EN"	ON as long motor runs then, Code #1, OFF Code #2, ON Code #3 OFF	ON after motor stops	Station light taken up to 4 seconds to come on after motor stops. Correct indications are Code #1 ON #2 OFF #3 ON
6. EN. Switch to "Off"	ON while motor runs	ON	
7. Repeat 2, 3, 4, 5 and 6 for each code on Figure 1			
8. Repeat steps 1 thru 7 for power settings of 29 VDC and 23 VDC			

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4.5.4 Environmental Tests. During or after subjection to the environmental tests, as specified in the test, the simulator shall be subjected to the functional tests. Failure to satisfy the requirements of the functional tests shall be considered as failure to pass the environmental test and cause for rejection.

4.5.4.1 High Temperature and Thermal Shock Test. The simulator shall be subjected to a temperature of +165°F for at least three hours. The functional tests shall be performed at this specific temperature with an operating voltage of 26 ± 3 volts DC. Immediately upon completion of the above tests, the simulator shall be exposed to a temperature of -65°F for at least one hour. Upon completion of the exposure and while at -65°F the simulator shall be subjected to the functional tests with an operating voltage of 26 ± 3 volts DC. Visual inspection shall reveal no damage that might ultimately result in malfunction.

NOTE: The test chamber may be opened to change switch position.

4.5.4.2 Low Temperature and Thermal Shock Test. The simulator shall be subjected to a temperature of -65°F for at least three hours. The functional tests will be performed at this specific temperature with an operating voltage of 26 ± 3 volts DC. Immediately upon completion of the above tests, the simulator shall be exposed to a temperature of 165°F for at least one hour. Upon completion of temperature exposure and while at the 165°F the simulator shall be subjected to the functional tests, with an operating voltage of 26 ± 3 volts DC. Visual inspection shall reveal no damage that might ultimately result in malfunction.

NOTE: The test chamber may be opened to change switch positions.

4.5.4.3 Humidity Test. The humidity test shall be performed in accordance with MIL-STD-810, Method 507. Within one hour after the conclusion of the test, the functional tests shall be performed.

4.5.4.4 Shock Test. The shock test shall be performed in accordance with MIL-STD-810, Method 516, Procedure 1, Figure 516-2, (a) and (c). At the conclusion of the test, the simulator shall satisfy the requirements of the functional tests.

4.5.4.5 Vibration Test. Vibration testing of the simulator shall be performed in accordance with MIL-STD-810(USAF), Equipment Class 1, Mounting Method A, Test Curve C of Figure 514-1, except that Curve C should be extended to 2000 cps and time schedule II of Table 514-11 should be used in place of time schedule I. The unit must satisfy the requirements of the functional test once during cycling and once during each resonance.

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4.5.5 Reliability Test. The simulator shall be subjected to a reliability test consisting of performance of the functional test repeated 350 times. Each performance of the functional test shall be considered one cycle. Conditions for this test shall be:

a. Vibration - One cycle 20 to 2000 to 20 hz using 0.036 inch double amplitude or 10g limits (the lesser of the two) and sweeping at one octave/minute repeated once per hour of the test.

b. On-Off cycling - On for 3 hours and 10 minutes and off for 50 minutes of each four hour period of testing.

c. Temperature - $25^{\circ}C \pm 5^{\circ}C$

d. Altitude - 0 to 3000 feet above mean sea level

e. Humidity - 90 percent or less

f. DC voltage 26 ± 3

Failure of any type prior to 175 cycles shall be cause for rejection. Failure after 175 cycles and before 350 cycles requiring adjustment by means other than nominal controls shall be cause for rejection.

5. PREPARATION FOR DELIVERY. Preparation of the simulator for delivery shall be in accordance with MIL-E-17555.

5.1 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

SIMULATOR, SWITCH BOX, BOMB AND ROCKET DISPENSER
SMU-64/A

6. NOTES

6.1 Intended Use. The SMU-64/A is intended for use in SUU-20/A dispensers to furnish electrical responses to using aircraft similar to those from nuclear weapons.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be performed by the procuring activity.

c. Levels of preservation, packaging and packing.

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6.2.1 Preproduction Samples. It is expected that the contract or purchase order will specify that a minimum of three units representative of the production simulator will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.3 Production Delivery. No production simulator should be delivered prior to full approval of the preproduction samples.

6.4 Definitions:

a. Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance Tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

Custodian
Air Force - 15

Preparing Activity
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Project No. 1190-F-004

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

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MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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